Project 2: Clustering Gene Expression Data

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Table 1: Clustering agreement with reference labels. Higher ARI is better.

Dataset	Method	ARI (inliers)	ARI (all genes)
CHO	K-means	**0.4412**	0.4412
CHO	Ward hierarchical	0.4174	0.4174
CHO	Gaussian mixture	0.3767	0.3767
Iyer	K-means	0.3801	0.3347
Iyer	Ward hierarchical	0.3790	0.3330
Iyer	Gaussian mixture	**0.3816**	0.3559

1 Data and Method

We clustered two gene expression datasets: CHO has 386 genes and 16 expression features; Iyer has 517 genes and 12 features. Each row also contains a gene ID and a reference cluster label. The label -1 marks an outlier. We used only expression features for clustering and standardized each feature to mean zero and unit variance.

We compared three methods. **K-means** assigns genes to centers and minimizes within-cluster squared distance. **Ward hierarchical clustering** repeatedly merges groups with the smallest increase in within-cluster variance. **Gaussian mixture** fits a probabilistic component to each group; we used diagonal covariance. The number of clusters was set to the number of reference groups other than -1: five for CHO and ten for Iyer. All genes were included when fitting. K-means used 30 starts, the mixture used 10 starts, and random seed 42 was used where applicable.

2 Evaluation

We used adjusted Rand index (ARI) to compare predicted clusters with the reference labels. An ARI of 1 means perfect agreement; a value near 0 means chance-level agreement. Our main ARI excludes genes labeled -1 at scoring time because these methods assign every gene to a cluster. We also report ARI with all genes included.

K-means scored highest on CHO. On Iyer, the Gaussian mixture scored highest, but its lead over K-means was only 0.0015, so the three methods performed similarly. The lower all-gene scores on Iyer show the effect of its labeled outliers. CHO has no -1 labels, so both score columns are the same.

3 Visualization

Figures 1 and 2 show the reference labels and the three clustering results on the same two-dimensional PCA projection. The clustering methods used all expression features; PCA was used only for display. The first two principal components explain 61.1% of CHO variance and 70.3% of Iyer variance. Colors and cluster numbers are arbitrary across panels, so we compare the shapes and overlap of groups rather than matching colors.

4 Conclusion

K-means was the strongest method for CHO and is simple and fast, but it favors compact clusters. Ward clustering gives a hierarchy and is deterministic, but early merges cannot be undone. A Gaussian mixture can model groups with different spreads, but its result depends on distribution assumptions and initialization. None of these methods marks outliers directly. The PCA plots help inspect group separation, while ARI gives the clearer comparison with the supplied labels.

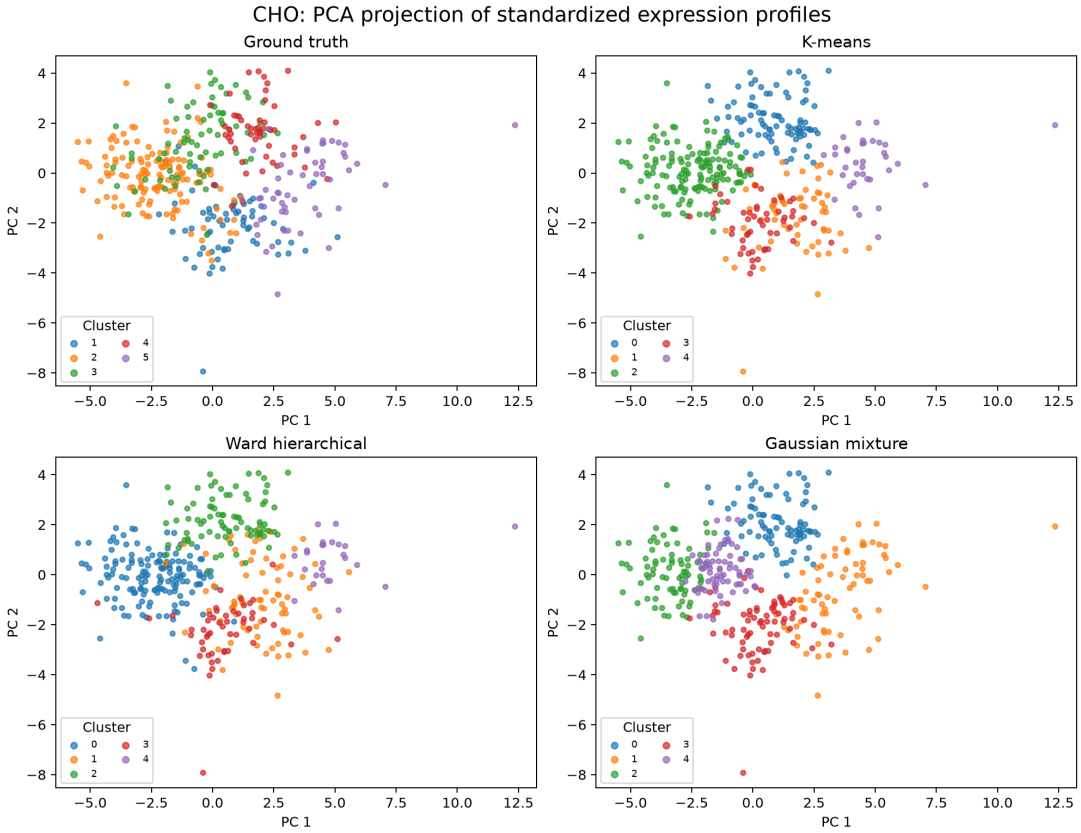


Figure 1: CHO: reference labels and clustering results in PCA space.

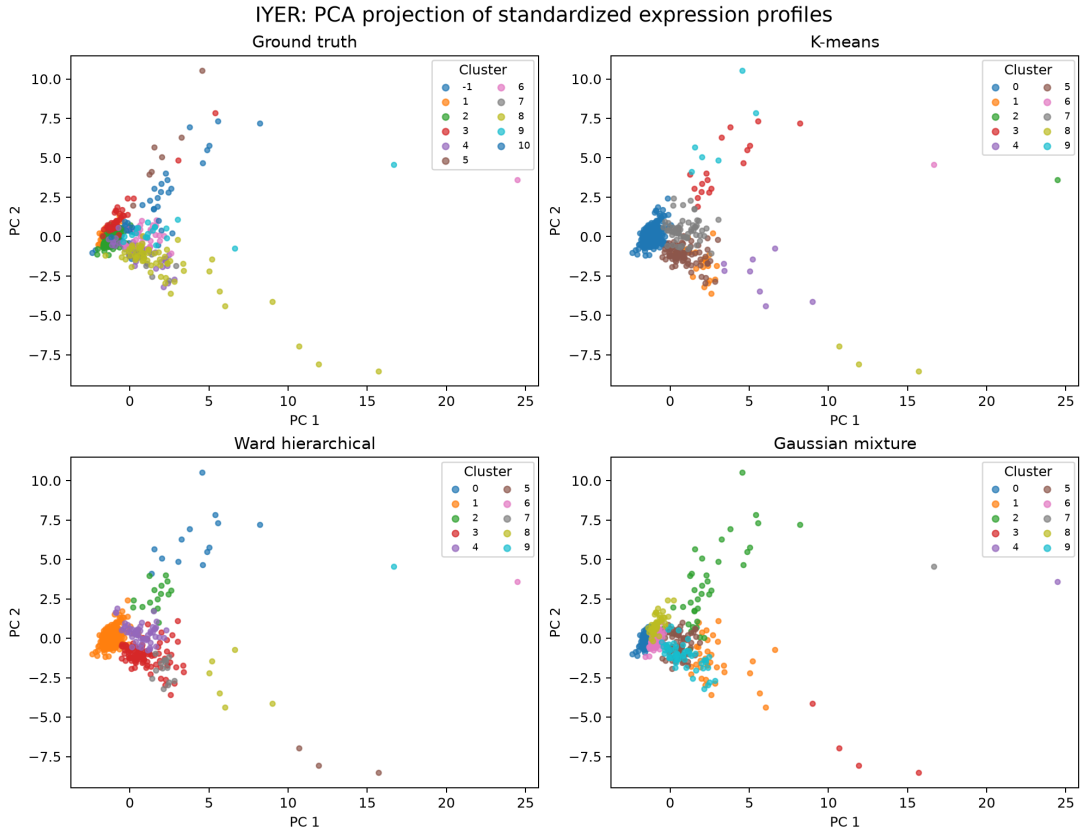


Figure 2: Iyer: reference labels and clustering results in PCA space.